DESIGN AND FABRICATION OF GEAR USING ALUMINIUM FOR IMPROVED MECHANICAL EFFICIENCY

MEB1204 - MANUFACTURING TECHNOLOGY- II

Submitted by

LARSHAD E 8115U24ME032

in partial fulfilment for the award of the degree of

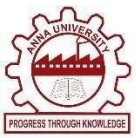
BACHELOR OF ENGINEERING

In

MECHANICAL ENGINEERING



K. RAMAKRISHNAN COLLEGE OF ENGINEERING



(AUTONOMOUS)

SAMAYAPURAM, TRICHY

ANNA UNIVERSITY

CHENNAI 600 025

DECEMBER-2025

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Under the Guidance of

Mr. B.VELUCHAMY., M.E., (Ph.D.)

Department of Mechanical Engineering

K. RAMAKRISHNAN COLLEGE OF ENGINEERING

MECHANICAL ENGINEERING

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Under

ANNA UNIVERSITY, CHENNAI

K. RAMAKRISHNAN COLLEGE OF ENGINEERING

(AUTONOMOUS)

Under

ANNA UNIVERSITY, CHENNAI

BONAFIDE CERTIFICATE

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DECLARATION BY THE CANDIDATE

I declare that to the best of my knowledge the work reported here in has been composed solely by myself and that it has not been in whole or in part in any previous application for a degree.

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SIGNATURE OF THE CANDIDATE

ACKNOWLEDGEMENT

I thank the Almighty God, for showing abundance of grace, without his blessings it would not have been possible for us to complete our project.

At this pleasing moment of having successfully completed our project, I wish to convey my sincere thanks and gratitude to our beloved kind Chairman, **Dr. K. RAMAKRISHNAN, B.E., Ph.D.,** who provided all the facilities to us.

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I extend our gratitude to all the teaching & non-teaching staff members of Mechanical Department, **K. RAMAKRISHNAN COLLEGE OF ENGINEERING**, for their kind help and valuable support to complete the project successfully.

I would like to thank our parents and friends for their constant support and encouragement throughout this project.

INSTITUTE VISION AND MISSION

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PO2 Problem analysis Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

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PO4 Conduct investigations of complex problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

PO5 Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

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PO9 Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO10 Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO11 Life-long learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context

ABSTRACT

The project titled “Design and Fabrication of gear using aluminium for improved mechanical efficeiency ” The demand for lightweight, corrosion-resistant, and cost-effective gears in automotive and machinery applications has driven the development of advanced aluminium-based composite components. This project focuses on the design, fabrication, and performance improvement of an aluminium gear reinforced with silicon dioxide (SiO_2) particles. The gear is manufactured using the sand-casting process, followed by precision machining to achieve accurate gear geometry, surface finish, and dimensional tolerances. In this work, aluminium was selected as the base matrix material due to its favourable strength-to-weight ratio, while SiO_2 particles were added as a reinforcement to enhance hardness, wear resistance, and overall mechanical stability. The study includes the preparation of the sand mould, incorporation of SiO_2 during melt processing, controlled casting, and post-casting machining operations such as turning, facing, and gear cutting. Mechanical properties such as hardness, tensile strength, surface roughness, and dimensional accuracy were evaluated and compared against that of a conventional unreinforced aluminium gear. The results indicate that SiO_2 reinforcement significantly improves the wear resistance and load-carrying capability of the cast gear without compromising machinability. Overall, this work demonstrates an effective approach for producing high-performance aluminium composite gears using an economical casting route, making it suitable for medium-load engineering applications.

KEYWORDS:

Gear shaping, dimensional tolerance, wear and tear, elongation

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CHAPTER 1
INTRODUCTION

Abstract :

The demand for lightweight, corrosion-resistant, and cost-effective gears in automotive and machinery applications has driven the development of advanced aluminium-based composite components. This project focuses on the design, fabrication, and performance improvement of an aluminium gear reinforced with silicon dioxide (SiO_2) particles. The gear is manufactured using the sand-casting process, followed by precision machining to achieve accurate gear geometry, surface finish, and dimensional tolerances. In this work, aluminium was selected as the base matrix material due to its favourable strength-to-weight ratio, while SiO_2 particles were added as a reinforcement to enhance hardness, wear resistance, and overall mechanical stability. The study includes the preparation of the sand mould, incorporation of SiO_2 during melt processing, controlled casting, and post-casting machining operations such as turning, facing, and gear cutting. Mechanical properties such as hardness, tensile strength, surface roughness, and dimensional accuracy were evaluated and compared against that of a conventional unreinforced aluminium gear. The results indicate that SiO_2 reinforcement significantly improves the wear resistance and load-carrying capability of the cast gear without compromising machinability. Overall, this work demonstrates an effective approach for producing high-performance aluminium composite gears using an economical casting route, making it suitable for medium-load engineering applications.

1.1 BACKGROUND OF GEAR MANUFACTURING

Gears are fundamental mechanical components that transmit power and motion between rotating shafts through interlocking teeth, enabling the efficient operation of machinery across industries such as automotive, aerospace, and industrial equipment. The history of gear manufacturing traces back to ancient civilizations, with the earliest known examples appearing in Greek and Roman machinery around 2700 BCE, where

wooden gears were used in chariots and water wheels. By 400 BCE, Aristotle referenced gears in his writings, highlighting their role in simple mechanisms like catapults and clocks. The transition to metal gears occurred during the medieval period, driven by the Industrial Revolution in the 18th and 19th centuries, which necessitated more durable materials for steam engines and early machinery. Mass production began in the 1800s with the advent of hobbing and shaping machines, evolving into precision engineering by the 20th century with the introduction of computer numerical control (CNC) systems.

Modern gear manufacturing encompasses a variety of processes, broadly categorized into soft and hard machining. Soft processes involve initial forming via hobbing, shaping, or milling on annealed blanks, followed by heat treatment and finishing operations like grinding or honing to achieve required tolerances. Hard processes, suitable for high-precision gears, include grinding directly on hardened blanks or advanced techniques like powder metallurgy and additive manufacturing. Materials selection is critical, influenced by factors such as load capacity, speed, and environmental conditions; common choices include low-carbon steels for general-purpose gears, alloy steels (e.g., AISI 8620) for high-strength applications, and non-ferrous alloys like bronze for corrosion resistance. Emerging trends incorporate composites and advanced alloys to address demands for lighter, more efficient components, setting the stage for innovations in gear design and fabrication.

1.2 NEED FOR LIGHTWEIGHT COMPOSITE GEARS

The escalating demand for energy efficiency, reduced emissions, and enhanced performance in automotive and machinery sectors has propelled the development of lightweight composite gears as a strategic solution. Traditional steel gears, while robust, contribute significantly to vehicle weight—accounting for up to 10-15% of drivetrain mass—leading to higher fuel consumption and inertial losses in transmissions. In the automotive industry, regulatory pressures like the EU's CO_2 emission targets (under 95 g/km by 2025) and the shift toward electric vehicles (EVs) underscore the urgency for weight reduction; lightweight gears can improve fuel

efficiency by 2-5% and extend EV range by minimizing unsprung mass.

Composite gears, integrating polymers, ceramics, or metal matrices with reinforcements, offer a compelling alternative by achieving 30-50% weight savings compared to metallic counterparts without sacrificing durability. For instance, carbon fiber-reinforced composites have demonstrated 40% mass reduction in gear bodies while maintaining torque transmission capabilities up to 500 Nm. In machinery applications, such as drones and portable tools, reduced inertia from composites lowers startup energy and noise levels, enhancing operational quietness by up to 10 dB. However, challenges like load-bearing under high stresses persist, with ongoing research focusing on hybrid designs to balance lightweighting with wear resistance. This need is particularly acute in EVs, where lightweight components optimize battery utilization and regenerative braking efficiency, driving innovations toward sustainable manufacturing.

1.3 ROLE OF ALUMINIUM AND SiO_2 ADDITIVES

Aluminium serves as an ideal matrix in metal matrix composites (MMCs) for gears due to its exceptional strength-to-weight ratio (density ~2.7 g/cm^3, roughly one-third of steel), superior corrosion resistance via a natural oxide layer, and excellent thermal conductivity, which aids in heat dissipation during high-speed meshing. In gear applications, aluminium alloys like A356 or 6061 provide a ductile base that facilitates casting and machining while supporting reinforcement integration for enhanced mechanical properties. When reinforced with silicon dioxide (SiO_2) particles, these MMCs exhibit synergistic improvements: SiO_2, with its high hardness (Mohs scale ~7) and thermal stability, acts as a dispersion strengthening agent, increasing wear resistance by 2-5 times through abrasive load transfer and reducing coefficient of friction.

In Al-SiO_2 composites, SiO_2 particles (typically 5-15 wt%) refine the microstructure, mitigating issues like porosity and improving tensile strength by 30-80% (from ~150 MPa to 200-280 MPa) and hardness by 20-50% (HV 60-80 to 90-120). This is particularly beneficial for gears under Hertzian contact stresses, where unreinforced

aluminium suffers from galling and fatigue. Studies on stir-cast Al-SiO_2 MMCs reveal enhanced creep resistance and fatigue life, attributed to SiO_2's role in pinning dislocations and stabilizing the matrix. Hybrid additions, such as SiO_2 with nano-graphene, further boost yield strength while preserving ductility (3-6% elongation), making these composites suitable for medium-load gears in corrosive environments like automotive transmissions.

1.4 METAL CASTING :

Metal casting is a versatile manufacturing process and one of the earliest metals shaping method known to human beings. It generally means pouring molten metal into a refractory mold with a cavity of the shape to be made and allowing it to solidify. When solidified, the desired metal object is taken out from the refractory mold either by breaking the mold or by taking apart. The solidified object is called casting. Fig 1.1 depicts casting process in foundry. Casting can produce products from few grams to several hundred tons and from simple shapes like watch cases to most complex parts like engine blocks. Almost any metal or alloy which can be easily melted is castable. Casting has many process variations depending upon the material, the type of pattern, mold and the pouring technique like sand casting, investment casting, die casting, squeeze casting and lost foam casting. Sand casting is the most widely used process which can be used to produce intricate parts in almost every metal that can be melted. According to worldwide census of casting production over 75 million metric tons of castings are produced annually. India is the fourth largest producer of castings. For successful production of castings, one needs knowledge in preparation of molds and patterns, melting and pouring of liquefied metal, solidification and further cooling to room temperature, inspection and quality control. Because of the complex physics involved and the number of steps needed to produce a casting, the parameters which govern the quality of a casting are huge in number. The major sources of defects arise from inappropriate design of the part, feeding and gating system.



Fig 1.1 Metal casting process in foundry

1.4.1 MOTIVATION FOR USING SAND CASTING

Sand casting remains a cornerstone in aluminium gear manufacturing due to its unmatched versatility, cost-effectiveness, and adaptability for complex geometries, making it an economical choice for prototypes and low-to-medium volume production. Unlike die casting, which requires expensive tooling, sand casting utilizes reusable silica-based molds that can be easily modified, reducing lead times to days and setup costs by up to 70% for intricate designs like gear teeth profiles. For aluminium, this process leverages the metal's low melting point (~660°C) and fluidity, enabling near-net-shape castings with minimal waste and high material utilization rates (>90%).

Key motivations include design flexibility—allowing undercuts and thin sections unattainable in forging—and compatibility with reinforcements like SiO_2, where stir-casting variants ensure uniform particle distribution via mechanical agitation. In gear production, sand casting facilitates the creation of rough blanks for subsequent machining (e.g., hobbing), achieving dimensional accuracies within IT8-IT10 grades. Environmentally, it aligns with sustainability goals through recyclable molds and lower energy demands compared to precision forging. Despite potential defects like porosity (mitigated by degassing), its scalability from small runs to large parts positions sand casting as a bridge between R&D and industrial adoption for lightweight Al-SiO_2 gears.



Fig 1.2 Sand casting process in foundry

1.5 PROBLEM STATEMENT

Conventional gear materials like steel excel in strength but impose penalties in weight, corrosion susceptibility, and manufacturing costs, limiting their viability in weight-critical applications such as electric drivetrains and portable machinery. Unreinforced aluminium gears address weight concerns but fall short in wear resistance and load capacity, exhibiting high friction coefficients (>0.3) and fatigue failure under cyclic stresses exceeding 200 MPa, leading to premature pitting and scoring in meshing contacts. The integration of ceramic reinforcements like SiO_2 in aluminium matrices promises enhanced performance, yet challenges persist in achieving uniform dispersion during casting, controlling porosity (<2%), and maintaining machinability without excessive tool wear.

This project addresses the core problem: developing a cost-effective, lightweight Al-SiO_2 composite gear that balances improved hardness (target >100 HV) and tensile strength (>250 MPa) with acceptable surface finish (Ra <1.6 μm) and dimensional tolerances (ISO 8), while overcoming sand casting limitations like particle agglomeration and thermal mismatch-induced cracks. Without such advancements, the automotive sector risks suboptimal fuel efficiency gains (potentially <3%) and higher lifecycle costs from frequent replacements in medium-load scenarios.

1.6 OBJECTIVES OF THE PROJECT

The primary aim of this project is to design, fabricate, and evaluate a SiO_2-reinforced aluminium gear using sand casting and precision machining to enhance mechanical performance for medium-load applications. Specific objectives include:

- To select and prepare an optimal aluminium alloy matrix (e.g., A356) with 5-10 wt% SiO_2 particles, ensuring uniform reinforcement distribution via stir-casting techniques.
- To develop and execute a sand casting process for gear blanks, incorporating mould design, melt processing, and defect minimization strategies.
- To perform post-casting machining operations (turning, facing, hobbing) to achieve precise gear geometry, surface integrity, and tolerances compliant with AGMA standards.
- To characterize mechanical properties—hardness, tensile strength, wear rate, and fatigue life—comparing the composite gear against unreinforced counterparts.
- To assess overall performance under simulated loads, validating improvements in wear resistance (>2x) and load-carrying capacity without compromising ductility or cost-effectiveness.

These objectives guide the fabrication of a prototype gear suitable for automotive transmissions, demonstrating a scalable route for composite integration.

1.7 SCOPE OF THE STUDY

This study encompasses the end-to-end development of a SiO_2-enhanced aluminium spur gear (module 2-3, 20-30 teeth) for medium-load applications (torque <500 Nm), focusing on sand casting as the primary fabrication method followed by CNC machining. It includes material synthesis via stir casting, mechanical testing per ASTM standards (e.g., E8 for tensile, E466 for fatigue), and wear evaluation using pin-on-disk setups. Comparisons are limited to unreinforced A356 aluminium, with SiO_2 loadings

up to 10 wt%; hybrid reinforcements or advanced simulations (e.g., FEA for stress distribution) are excluded. The scope is confined to laboratory-scale prototypes, excluding full-scale field trials, cost modeling beyond raw materials, or environmental impact assessments. Findings apply to non-high-speed gears in corrosive-free environments, providing foundational insights for industrial scaling.

CHAPTER 2
LITERATURE SURVEY

1. **Ravi, B., Joshi, D., 2007. Feed ability analysis and optimization driven by casting simulation. Indian Foundry J. 53 (6),71—78.**

   This article mainly focussed on design and optimization technique based on casting related defects and their research and outcomes. Ravi and Joshi worked on computer aided casting design and simulation of feeder and gating design of castings using Auto-CAST software and they describe how it assists in designing, modelling, simulating, analyzing and improving cast products. From their results it is concluded that the casting simulation helps visualize mold filling and casting solidification. It predicts related defects like shrinkage, porosity and hard spots and optimize the casting design to achieve the desired quality with high yield.

2. **Hassan, I., Sheikh, A.K., Al-Yusuf, A., 2012. Mold design optimization for sand casting of complex geometries using advance simulation tools, material manufacturing Process.**

   Hassan investigated the impeller shaped casting using MAGMASOFT Software. The effect of riser and gates on parameters, such as filling pattern, pressure and speed, cooling rate, solidification and related defects, was studied. The results of their studies shows that use of this casting simulation will keep casting producers competitive & profitable and with this simulation software the casting process from filling to solidification is simulated. Foundries relying on this simulation tool can have the advantage of the latest casting process design technology that calculates residual stresses, metallurgical prediction improves quality, reduces lead times and cost.

3. **Choudhari, C.M.Padalkar, K.J., Duhamel, K.K.Narkhede, B.E.Mahajan, S.K., 2013. Defect free casting by using simulation software. Appl. Mechanical Matter.**

   Choudhari suggested that by optimization method casting related defects can be improved. The proper location, size and design of gating and feeder system using simulation technology improved the shrinkage porosity and cracks in casting. From the results of their studies it is clear that the defects in casting process can be reduced to some extent using simulation software.

4. **B. Ravi, "Casting Simulation and Optimisation: Benefits, Bottlenecks, and Best Practices," Indian Foundry Journal, 54(1), 47-52,2008.**

   From this literature B.Ravi concluded that the bottlenecks and non- value added time in casting development can be minimized by adopting CAD, 10 Intelligent methoding and simulation technologies. These have been developed, successfully demonstrated on industrial castings, and now being used in several organizations. Several innovative algorithms, including VEM, geometric reasoning, and automatic solid modelling dramatically compressed the iteration time for methoding modification and simulation to less than one hour for even complex castings. Further, the simple and logical user interface greatly improved the learning curve for engineers, to just a few hours. As a result, even small foundries with little or no previous exposure to CAD/CAM software are able to effectively use the program to improve their casting quality, yield and productivity. It has also proven to be very useful for verifying the manufacturability of a casting and improving it by minor modifications to part geometry, before freezing the design. In future, it will be possible to offer automated methoding and simulation functionality over the Internet, enabling access to this technology to even SME foundries in remote areas. This study has been very helpful for understanding the best practices of simulation technology in manufacturing industry.

5. **Ravi B, Metal Casting: Computer-Aided Design and Analysis, PHI, New Delhi, ISBN 81 203 2726 8, 4th print, 2008.**

B.Ravi concluded that in recent years, computer simulation of casting solidification has gained much ground, owing to the constant and painstaking efforts of researchers to make such software tools more reliable and easy to use. A significant number of real life case studies are also available in technical journals and proceedings of conferences related to casting. Still, only a handful of aluminium foundries are using these software tools today. This is owing to several challenges posed by first-time users. We first present an overview of computer-aided methods design. This is followed by the difficulties faced by foundry users, and how to overcome them through best practices gleaned from our experience with several simulation projects and consultants. From this he stated the need for integration of CAD with manufacturing industries for their rapid development.

6. **Gear manufacturing methods – hadhav ajith,dhamdhere supriya.**

Gear is an important element of mechanical power transmission. It is manufactured by many processes such as casting, milling, hobbing & shaping. Among this manufacturing milling is preferred for job production & hobbing is preferred for mass production. The gear manufacturing operations mostly perform on milling machine & CNC milling machines. In modern manufacturing high dimensional accuracy & fine surface finish play an important role. The grinding process is commonly used as finishing operation. The all gear manufacturing has its own importance. The main purpose of this paper is to give the literature survey on various gear manufacturing methods.

7. **Study on optimization of spur gear performance with titanium carbide incorporated aluminum matrix composite, R.siva, K.A.Vimalson, p.Yogeskumar**

In this paper, the mechanical properties and surface morphology of Aluminium alloy 6061 composite with differing weight percentages of 40 µm grain size Titanium carbide used to fabricate spur gears are studied. A comparison is made between the conventional spur gear model and the developed composite with better strength, the complete analysis is done using ANSYS. The Aluminium alloy 6061 composite samples reinforced with 5%, 10% and 15% by weight Titanium carbide particles are fabricated by stir casting process. Tensile, hardness and impact tests are conducted for all the composites to determine their mechanical properties. The modelling and Finite Element Analysis of the spur gear is done using CATIA and ANSYS respectively. It is noted that increased weight percentage of Titanium carbide gives higher tensile strength, impact strength and hardness values. ANSYS analysis of spur gear of aluminium alloy 6061 with 15% Titanium carbide composite shows that it has better fatigue life than the gear made out of conventional cast iron

CHAPTER-3
PROBLEM IDENTIFICATION

3.1 RELATED STUDY - DEFECTS IN CASTING PROCESS

Casting is a process which carries risk of failure occurrence during all the process of accomplishment of the finished product. Hence necessary action should be taken while manufacturing of cast product so that defect free parts are obtained. Mostly casting defects are concerned with process parameters. Hence one has to control the process parameter to achieve zero defect parts. For controlling process parameter one must have knowledge about effect of process parameter on casting and their influence on defect. To obtain this all knowledge about casting defect, their causes, and defect remedies one has to be analyze casting defects. Casting defect analysis is the process of finding root causes of occurrence of defects in the rejection of casting and taking necessary step to reduce the defects and to improve the casting yield.

During the process of casting, there is always a chance where defect will occur. Minor defect can be adjusted easily but high rejected rates could lead to significant change at high cost. Therefore it is essential for die caster to have knowledge on the type of defect and be able to identify the exact root cause, and their remedies.

- Casting defects can be classified as follows
- Filling related defect
- Shape related defect
- Thermal defect
- Defect by appearance

2.1.1 FILLING RELATED DEFECTS

- Blow holes

Blow hole is a kind of cavities defect, which is also divided into pinhole and subsurface blowhole. Pinhole is very tiny hole. Subsurface blowhole only can be seen after machining. Gases entrapped by solidifying metal on the surface of the casting as shown in fig 3.1, this result in a rounded or oval blowhole as a cavity and frequently associated with Slag or oxides. The defects are nearly always located in the cope part of the mold in poorly vented pockets and undercuts.

Remedies



Fig 3.1 Blow holes in casting

Reduce moisture content of sand. Improve conditioning of the sand. Reduce inert dust content. Improve gas permeability. Endeavour to use coarser sand. Reduce bentonite and carbon carrier content. Reduce sand temperature. Install a sand cooler if necessary. Increase sand quantity. Reduce bentonite content or use bentonite with a high specific binding capacity and good thermal stability. 12

Sand burning

Burning-on defect is also called as sand burning, which includes chemical burn-on, and metal penetration, thin sand crusts firmly adhering to the casting. The defect occurs to a greater extent in the case of thick walled castings and at high

temperatures. The high temperature to which the sand is subjected causes sintering of the bentonite and silicate components. In addition, the always present iron oxides combine with the low-melting-point silicates to form iron silicates, thereby further reducing the sinter point of the sand. Sintering and melting of the impurities in the molding sand enable the molten iron to penetrate even faster, these layers then frequently and firmly adhering to the casting surface.

Remedies

Even out incoming metal flow. Reduce pouring rate. Reduce liquid metal temperature

Sand inclusion

Sand inclusion and slag inclusion are also called as scab or blacking scab. They are inclusion defects and looks like there is slag inside of metal castings. Irregularly formed sand inclusions, close to the casting surface, combined with metallic protuberances at other points as shown in fig 2.2. Sand inclusion is one of the most frequent causes of casting rejection. It is often difficult to diagnose, as these defects generally occur at widely varying positions and are therefore very difficult to attribute to a local cause. Areas of sand are often torn away by the metal stream and then float to the surface of the casting because they cannot be wetted by the molten metal. Sand inclusions frequently appear in association with CO blowholes and slag particles. Sand inclusions can also be trapped under the casting surface in combination with metal oxides and slag's, and only become visible during machining. If a loose section of sand is washed away from one part of the mold, metallic protuberances will occur here and have to be removed.



Fig 3.2 Sand inclusion defect 13

Remedies

Avoid high pouring rates and impact of metal stream against mold walls.

Shorten pouring times, improve distribution of gates

 Cold lap or cold shut

Cold lap or also called as cold shut and is caused when two streams do not fuse together properly thus forming a discontinuity in casting as shown in fig 2.3. It is a crack with round edges. Cold lap is because of low melting temperature or poor gating system.

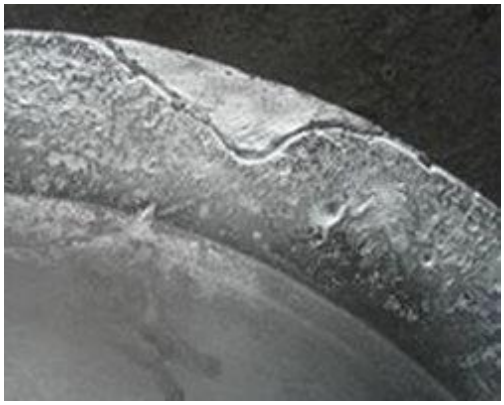


Fig 3.3 Cold shuts formed in casted object

Remedies

Adjust proper pouring temperature. Modify design of gating system

 Misrun

Misrun defect is a kind of incomplete casting defect, which causes the casting uncompleted. The edge of defect is round and smooth as shown in fig 2.4. When the

metal is unable to fill the mold cavity completely and thus leaving unfilled portion called misrun.



Fig 3.4 Misrun defect in casting 14

Remedies

Adjust proper pouring temperature. Modify design of gating system.

 Gas porosity

The gas can be from trapped air, hydrogen dissolved in aluminium alloys, moisture from water based die lubricants or steam from cracked cooling lines. Air is present in the cavity before the shot. It can easily be trapped as the metal starts to fill the cavity. The air is then compressed as more and more metal streams into the cavity and the pressure rises. When the cavity is full it becomes dispersed as small spheres of high pressure air. Fig 2.5 depicts the gas porosity defect in casting.

Remedies

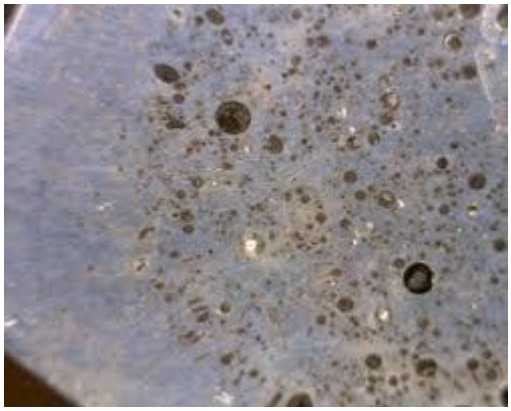


Fig 3.5 Gas porosity defect in casting

Increase metal pouring temperature. Pour metal as rapidly as possible without interruption. Improve mold filling by modification to running and gating system. Reduce gas pressure in the mold by appropriate adjustment to molding material properties and ensuring Adequate venting of molds and cores.

3.1.2 SHAPE RELATED DEFECTS

- **Mismatch defect**

Mismatch in mold defect is because of the shifting molding flaks. It will cause the dislocation at the parting line. A mismatch is caused by the cope and drag parts of the mold not remaining in their proper position. This is caused by loose box pins, inaccurate pattern dowel pins or carelessness in placing the cope on the drag. Fig 2.6 shows mismatch defect in casting.

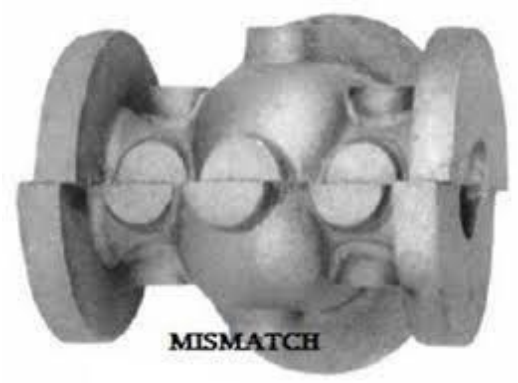


Fig 3.6 Mismatch defect in casting15

Remedies

Proper molding box and closing pins should be used

- **Flash defect**

Flash can be described as any unwanted, excess metal which comes out and attached to the cavity or runner. Typically it forms a thin sheet of metal at the parting faces as shown in fig 3.7. There are a number of different causes of flash and the amount and severity can vary from a minor inconvenience to a major quality issue. At the very least, flash is waste material, which mainly turns into dross when re-melted, and therefore is a hidden cost to the business.

Remedies



Fig 3.7 Flash defect in casting

If your sprue is very tall and the casting covers a wide area of the mold face, it's very possible for the mold to actually be forced up by the hydrostatic pressure of the metal. Proper design of sprue helps as remedy.

2.1.3 THERMAL DEFECTS

 Cracks or tears

Cracks can appear in castings from a number of causes. Some cracks are very obvious and can easily be seen with the naked eye. Other cracks are very difficult to see without magnification. Excessive porosity in critical regions of the part may be the reason and fig 2.8 depicts the cracks formed during casting.



Fig 3.8 Cracks or tears formed during casting 16

Remedies

Reduce pouring temperature. Avoid superheating of metal. Use chills. Provide feeders Avoid early knockout and give sufficient cooling time. Reduce sharp corners in gating design.

 Shrinkage

Shrinkage defects occur when feed metal is not available to compensate for shrinkage as the metal solidifies. Shrinkage defects can be split into two different types: open shrinkage defects and closed shrinkage defects. Open shrinkage defects are open to the atmosphere, therefore as the shrinkage cavity forms air compensates. There are two types of open air defects: pipes and caved surfaces. Pipes form at the surface of the casting and burrow into the casting, while caved surfaces are shallow cavities that form across the surface of the casting. Closed shrinkage defects also known as shrinkage porosity, are defects that form within the casting. The isolated pools of liquid form inside solidified metal, which are called hot spots. The shrinkage defect usually forms at the top of the hot spots. They require nucleation point, so impurities and dissolved gas can induce closed shrinkage defects as shown in fig 3.9. The defects are broken up into macro porosity and micro porosity, where macro porosity can be seen by the naked eye and micro porosity cannot.



Fig 3.9 Shrinkage defect in casting process

Remedies

The general technique for eliminating shrinkage porosity is to ensure that liquid metal under pressure continues to flow into the voids as they form.

3.1.4 DEFECTS BY APPEARANCE

 Cavities

Blowholes, pinholes, Smooth-walled cavities, essentially spherical, often not contacting the external Casting surface (Blowholes). The defect can appear in all regions of the casting. Blowholes and pinholes are produced because of gas entrapped in the metal during the course of solidification 17

Remedies

Make adequate provision for evacuation of air and gas from the mold cavity, Increase permeability of mold and cores.

 Incomplete casting

Poured short and portion of the casting is missing. The edges adjacent to the missing section are slightly rounded and all other contours conform to the pattern. The sprue, risers and vents are filled only to the same height above the parting line, as is the casting. Insufficient quantity of liquid metal in the ladle and premature interruption of pouring due to workman's error may be the causes.

Remedies

Have sufficient metal in the ladle to fill the mold, Check the gating system.

Instruct pouring crew and supervise pouring practice.

 Defective surface

Flow marks on the surfaces of castings. The defect appears as lines which trace the flow of the streams of liquid metal.

Remedies

Increase mold temperature. Lower the pouring temperature. Modification of gate size and location serves as remedy.

3.2 SUMMARY OF LITERATURE

With rapid development in computer hardware and software technologies, the usage of simulation tools in foundries has increased. casting process simulation is now a well established tool for the optimisation of casting design as well as process design .They have reached a point where the simulation tools have now become an integrated part of entire casting production process, from casting design to adjusting production parameters. There are many aspects of casting process that can be evaluated by using casting simulation process. As seen earlier, the casting process is extremely complex, and analytical procedures are very difficult to apply without using large number of assumptions many of the assumptions though simplify the calculation procedures, do not guarantee accurate results , this calls for some trial and error method to achieve sound casting which is time consuming and expensive. 18

Casting simulation utilises finite difference and finite element procedures depending upon the complexity of the problem. It provides large range of results of the complete behaviour of casting during filling as well as solidification. The simulation process allows establishing the appropriate casting design specific process parameters to control manufacturing process. Using the casting simulation as a virtual test in foundry it is possible to do a number of what-if scenarios to establish the optimum casting design and foundry process and very effective for obtaining good quality and defect free casting.

CHAPTER 4

OBJECTIVE AND SCOPE OF THE PROJECT

The primary objective of this project is to design, fabricate, and evaluate an aluminium gear reinforced with SiO_2 particles to enhance its mechanical performance. The project aims to develop a lightweight yet high-strength composite gear that offers improved hardness, wear resistance, and load-carrying capability compared to conventional aluminium gears. To achieve this, the project focuses on optimizing the material composition, ensuring uniform dispersion of SiO_2 in the aluminium matrix through stir casting, and producing a defect-free composite blank suitable for precision machining. Another key objective is to accurately machine and cut the gear teeth using appropriate processes such as turning, milling, and hobbing, ensuring dimensional accuracy and proper tooth geometry. The project also aims to assess the mechanical properties of the composite gear through hardness, tensile, wear, and microstructural tests, enabling a comparison with standard aluminium or steel gears. Ultimately, this project seeks to demonstrate the feasibility of using aluminium–SiO_2 composites as a cost-effective and efficient alternative for medium-load gear applications.

The scope of this project covers the entire process required to produce an aluminium–SiO_2 composite gear, starting from material selection to performance evaluation. It includes the selection of a suitable aluminium alloy and SiO_2 reinforcement, preparation of raw materials, and fabrication of the composite using the stir casting method. The scope further extends to machining operations needed to convert the cast composite into a precise gear blank, followed by cutting the gear teeth using hobbing or shaping techniques. Additionally, the project involves conducting mechanical testing to evaluate hardness, tensile strength, wear resistance, and microstructural characteristics of the composite. The data obtained will be used to compare the composite gear's performance with conventional gear materials. Documentation, analysis, and interpretation of results

also fall within the project's scope, along with identifying potential improvements, limitations, and applications. Overall, the project encompasses design, casting, machining, testing, and evaluation activities needed to validate the use of aluminium–SiO_2 composites in gear manufacturing.

CHAPTER-5

MATERALS AND METHODS

5.1 Electrical material and components:

Table 5.1

ITEM	SPECIFICATION / NOTES
ALUMINIUM ALLOY	LM6 depending on the required strength
SiO_2 Reinforcement Particles	5–12 wt.% (particle size 20–80 μm)
Flux / Wetting Agent	Magnesium (0.5–1%) to improve SiO_2 wettability
Mold Material	Cast iron / mild steel / sand mold depending on casting method
Lubricant / Cutting Fluid	For machining operations (CNC machining)

1. **LM6**

   LM6 is a high-silicon aluminium alloy known for excellent castability and corrosion resistance. It flows well in molds and is ideal for producing complex shapes like gear blanks. When reinforced with SiO_2 particles, its strength and wear resistance improve significantly.

2. SiO_2 Reinforcement Particles

Silicon dioxide (SiO_2) particles are added in the range of 5–12 wt.% to form a metal matrix composite. These hard ceramic particles increase hardness, improve wear resistance, and enhance load-bearing capacity. Particle sizes of 20–80 μm ensure good distribution and bonding within the aluminium matrix.

3. Flux / Wetting Agent (Magnesium)

A small amount of magnesium (0.5–1%) is added to improve the wettability of SiO_2 particles within molten aluminium. It helps the reinforcement mix uniformly by reducing surface tension and preventing particle clustering during stir casting.

4. Mold Material

Depending on the casting method, cast iron, mild steel, or sand molds are used. Sand molds are most common due to low cost and design flexibility. Metal molds provide better surface finish and dimensional accuracy, useful for producing consistent gear blanks.

5. Lubricant / Cutting Fluid

During machining operations like turning, milling, and lubricants or cutting fluids reduce friction and heat. They improve tool life, help achieve smooth surface finish on the aluminium gear, and prevent material wear or thermal distortion.

5.2 Mechanical material and components:

COMPONENT	DETAILS
Sand moulding	For the aluminium to be casted
Mechanical Stirrer	Graphite-coated impeller, controlled RPM (300–600 rpm)
Preheating Furnace	For preheating SiO_2 particles (400–500°C)
Lathe Machine	For turning and facing the cast composite blank
Gear Hobbing / Shaping Machine	For cutting gear teeth (spur/helical)
Hardness Testing Machine	Brinell
Tensile Testing Machine	For composite specimen testing
Vernier Caliper & Micrometer	Dimensional measurement tools

1. Sand Moulding

Sand moulding is used to create the cavity where molten aluminium composite is poured. It is cost-effective, easy to shape, and suitable for making gear blanks with good dimensional accuracy.

2. Mechanical Stirrer

A mechanical stirrer with a **graphite-coated impeller** is used in stir casting to disperse SiO_2 particles uniformly in molten aluminium. Controlled speeds of **300–600 rpm** ensure proper vortex formation and homogeneous mixing.

3. Preheating Furnace

SiO_2 particles are preheated to **400–500°C** before mixing. This removes moisture, improves bonding, and prevents thermal shock when added to molten aluminium, ensuring better wettability and uniform distribution.

4. Lathe Machine

The lathe is used for initial machining operations such as **turning, facing, and sizing** the cast gear blank. This step ensures concentricity and prepares the blank for further machining.

6. Gear Shaping Machine

This machine is used to cut the **gear teeth profile** (spur or helical). Shaping is common due to accuracy and speed, while shaping is used for internal or special tooth profiles.

7. Hardness Testing Machine (Brinell)

The Brinell hardness tester measures the surface hardness of the aluminium–SiO_2 composite. It helps evaluate wear resistance and the strengthening effect of the reinforcement.

8. Tensile Testing Machine

Used to test standard composite specimens to measure **ultimate tensile strength, yield strength, and elongation**. It shows how reinforcement affects the mechanical behaviour of the composite.

9. Vernier Caliper & Micrometer

These instruments are used for precise **dimensional measurement**. They ensure accuracy in thickness, diameter, and gear tooth dimensions during and after machining.

Method explanantion:

The fabrication of the SiO_2-reinforced aluminium gear begins with the preparation of a sand mould shaped according to the required gear blank dimensions, including proper gating and riser systems to ensure smooth metal flow during casting. LM6 aluminium alloy is then melted in a crucible furnace at around 680–720°C, while SiO_2 reinforcement particles are preheated to 400–500°C in a preheating furnace to remove moisture and improve bonding with the molten metal. Once the aluminium is fully molten, a mechanical stirrer with a graphite-coated impeller is used to create a controlled vortex at 300–600 rpm, into which the preheated SiO_2 particles are gradually added along with a small amount of magnesium to enhance wettability. The mixture is stirred thoroughly to ensure uniform dispersion of particles and then poured carefully into the sand mould for solidification. After the composite gear blank cools, it is removed and subjected to initial machining on a lathe machine for turning, facing, and sizing operations to achieve accurate outer diameter, bore alignment, and thickness. Further machining is carried out on a milling machine for center drilling, surface finishing, and keyway preparation, followed by precise gear tooth cutting using a gear hobbing or shaping machine to obtain the desired spur or helical profile. The finished gear is deburred, cleaned, and inspected before being tested mechanically, where Brinell hardness testing and tensile testing are performed to evaluate the composite's hardness, strength, and overall performance. Finally, dimensional accuracy is verified using Vernier calipers and micrometers to ensure the gear meets the required specifications and tolerances.

CHAPTER 6

SELECTION OF MATERIALS

The selection of materials plays a crucial role in ensuring the performance, durability, and reliability of the fabricated gear. Based on the requirements of mechanical strength, castability, machinability, and wear resistance, this project uses a combination of aluminium alloy and SiO_2 reinforcement to develop a high-performance metal matrix composite suitable for medium-load gear applications. The primary matrix material selected is **Aluminium Alloy LM6 (Al–Si12)**, as referenced in the document, due to its excellent fluidity, corrosion resistance, and suitability for complex castings such as gear blanks

LM6 provides a lightweight and easily castable base with good dimensional accuracy and low shrinkage during solidification, making it ideal for sand casting operations.

To enhance the mechanical performance of LM6, **Silicon dioxide (SiO_2) particles** are chosen as the reinforcement material. SiO_2 is selected because of its high hardness, thermal stability, and ability to significantly improve wear resistance when dispersed in an aluminium matrix. The PDF indicates the use of particulate reinforcements in similar composite fabrication processes, supporting this choice for improved strength and surface durability gear

. The selected reinforcement percentage (5–12 wt.%) provides a balanced combination of improved hardness and machinability. Furthermore, the particle size range of 20–80 µm ensures effective bonding and uniform distribution during stir casting, preventing agglomeration and ensuring enhanced load-bearing capability.

Additional auxiliary materials are selected to support the casting and machining process. **Magnesium (0.5–1%)** is used as a wetting agent to increase the affinity

between molten aluminium and SiO_2 particles, improving dispersion and reducing porosity. The casting is carried out using a **sand mould**, which is highlighted in the PDF as a suitable moulding method for aluminium composite components due to its low cost, thermal stability, and adaptability to different geometries. For machining, standard cutting fluids and lubricants are used to minimize tool wear during turning, milling, and gear hobbing operations.

Overall, the selected combination of LM6 aluminium alloy, SiO_2 reinforcement, magnesium wetting agent, and sand mould casting ensures the development of a composite gear with improved hardness, better wear resistance, and adequate strength for medium-load applications. The choices are consistent with the materials and processes referenced in the source document and align with the engineering requirements of the project.

CHAPTER 7
WORKING PRINCIPLE

7.1 Principle:

The principle behind manufacturing the SiO_2-reinforced aluminium gear is based on **forming a metal matrix composite (MMC) through stir casting**, where aluminium alloy (LM6) acts as the matrix and SiO_2 particles serve as the reinforcement. When the aluminium is melted and stirred at a controlled speed, a vortex is created that allows the preheated SiO_2 particles to disperse uniformly throughout the molten metal. With the help of a wetting agent like magnesium, the particles bond effectively with the aluminium, forming a strong and homogeneous composite. After solidification in a sand mould, the composite blank is machined using conventional processes such as turning, milling, and gear hobbing to achieve the final gear profile. The reinforced composite gear works on the principle of **enhanced strength, hardness, and wear resistance** due to the uniform distribution of SiO_2 within the aluminium matrix.

7.2 working principle:

The working principle of manufacturing a SiO_2-reinforced aluminium composite gear is based on the concept of **metal matrix composite (MMC) formation through stir casting**, followed by **conventional machining and gear tooth generation** to obtain the final component. In this process, LM6 aluminium alloy acts as the primary matrix material, while SiO_2 particles serve as the reinforcement that enhances strength, hardness, and wear resistance. The principle begins with melting the aluminium alloy and preheating the SiO_2 particles to remove moisture and thermal shock, ensuring proper bonding. A mechanical stirrer creates a controlled vortex inside the molten metal, allowing the SiO_2 particles to be added gradually. The stirring action, combined with the use of a wetting agent like magnesium, ensures uniform dispersion of the

particles within the aluminium matrix. This forms a homogeneous composite melt,

which is then poured into a sand mould where it solidifies into the shape of a gear blank.

Once solidified, the composite blank is subjected to a series of machining operations that convert it into the final gear. The working principle of machining involves removing excess material to achieve required dimensions, ensuring concentricity, and preparing the surfaces for accurate tooth cutting. Gear teeth are generated using gear hobbing or shaping, where a cutter progressively removes material until the desired profile, module, and number of teeth are achieved. The final gear works on the principle of **power transmission through meshing teeth**, where the reinforced aluminium composite provides improved mechanical strength and enhanced resistance to wear during operation.

Thus, the overall working principle integrates **composite material formation**, **controlled casting**, and **precision machining**, resulting in a lightweight, durable, and wear-resistant aluminium–SiO_2 composite gear suitable for medium-load mechanical applications.

CHAPTER 8

PROCESSS INVOLVED IN FABRICATION

1. Pattern & tooling preparation

- Prepare a 1:1.02 wooden (or 3D-printed) pattern for the gear blank with draft (≈1°), core prints for bore, and shrinkage allowance.
- Make gating, sprue and riser design on the pattern (sprue Ø ≈ 15 mm; riser volume ≈ 20% of casting).
- Inspect pattern surface finish (target Ra ≈ 3.2 μm) and install alignment/dowel pins in the flask

2. Sand mould preparation

- Mix green sand: silica sand (70–85%) + bentonite clay (≈4%) + water (≈2–3%) + organic additives (1–3% if needed).
- Ram sand around pattern (cope & drag), place vents (2 mm) and chaplets; apply parting agent.
- Measure mold properties: green hardness ≈ 60–70, permeability 20–30. Adjust if required.
- Assemble mold and check alignment; label gating/riser locations

3. SiO_2 pre-treatment

- Dry/preheat SiO_2 particles at 400–500°C for 30–60 min to remove moisture and volatiles.
- Weigh reinforcement for target loading (typical 5–12 wt%; project uses 8 wt% as optimized)

4. **Charge & melt aluminium**

- Charge LM6 / A356 alloy ingots into clay-graphite crucible.
- Heat to melt (target melt temperature ≈ 720–750°C — ~50°C superheat above liquidus).
- Cover melt with inert gas (argon) or use a flux to reduce surface oxides

5. **Add wetting agent**

- Add magnesium (0.5–1 wt%) to the melt to improve wettability of SiO_2 and reduce particle rejection/clustering.
- Allow Mg to dissolve and homogenize briefly before stirring.

6. **Stir casting (reinforcement incorporation)**

- Insert mechanical stirrer with graphite-coated impeller; create stable vortex.
- Stir at controlled RPM: **300–600 rpm** (project uses ~500 rpm) to form vortex.
- Gradually add preheated SiO_2 particles into the vortex over 1–2 minutes to avoid splash and agglomeration.
- Continue stirring for ~5–10 minutes to ensure uniform dispersion (monitor for vortex stability and torque).
- Record melt temperature (keep ~720°C); avoid excessive oxidation.

7. **Final degas / skim**

After stirring, degas again briefly and skim off any remaining dross.

Check viscosity/flow — ensure no visible clusters.

8. Pouring into mould

- Pour using bottom-pour gating (if available) to minimize turbulence; keep pouring velocity low (<0.5 m/s).
- Maintain melt temperature during pour; fill mold steadily.
- Note time of pour and record thermocouple data (for process log).

9. Solidification & cooling

- Allow controlled cooling. Use insulating coatings (graphite/kaolin) to control cooling rate (~1–3°C/s through mushy zone).
- Monitor using thermocouples; allow to reach <100°C before shakeout.

10. Demoulding & fettling

- Break sand mold and remove casting.
- Remove sprue/riser with saw or chisel; grind gate areas flush; sandblast to clean surface.
- Inspect for visible defects (cracks, blowholes, sand inclusions). If suspected, perform ultrasonic or X-ray NDT.

11. Turning (lathe) — rough & finish

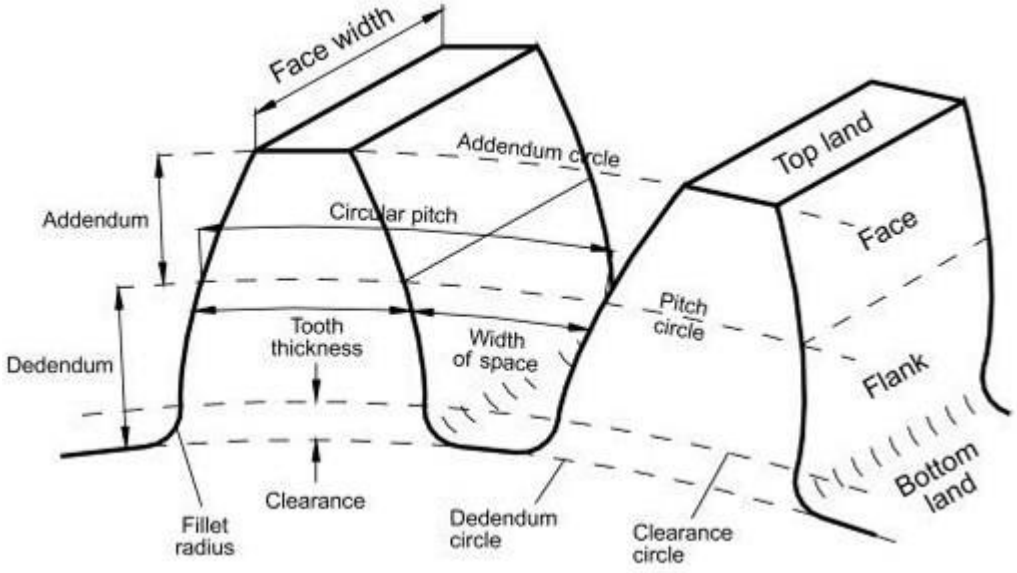
- Rough turn to remove casting skin and bring OD close to nominal (allow machining allowance ~0.5–1 mm).
- Cutting parameters (example): rough: Vc ≈150 m/min, N ≈1200 rpm, f ≈0.2 mm/rev, a_p ≈1.0 mm. Finish: Vc ≈180 m/min, f ≈0.08 mm/rev, a_p ≈0.2 mm to achieve Ra ≈1.2–1.6 μm.
- Use carbide inserts (coated for abrasive SiO_2). Apply MQL or light coolant as needed.

12. Finishing operations

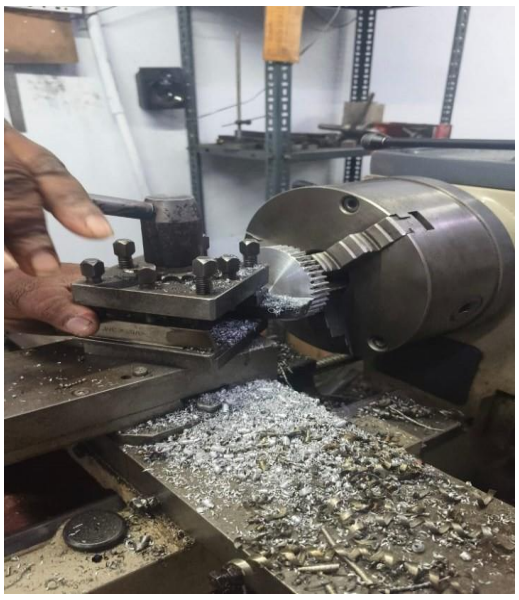
- Profile grind/hone teeth (if needed) to improve surface finish and contact fatigue life (target Ra ≤ 0.8 μm on tooth flanks).
- Deburr (vibratory or manual) to remove sharp edges and avoid stress raisers.
- Polishing/electrolytic polishing optional for corrosion resistance.

DESIGN AND TESTING OF THE GEAR

DESIGN OF THE GEAR:



FABRICATION OF GEAR:





KEY GEAR DIMENSIONS

Parameter	Symbol	Value (mm)	Formula
Pitch Diameter	d	40	$m \times Z$
Outside Diameter	D_o	44	$d + 2m$
Root Diameter	d_r	35	$d - 2.5m$
Face Width	b	12	$6m$
Center Distance	a_w	60	$(d_1 + d_2)/2$

MATERIALS REQUIRED

composition of green sand	percentage	reason
silica sand	70-85%	to provide reactiveness
clay	10-20%	to act as binder, imparts tensile and shear strength to the mould
water	3%	to activate the clay and provide elasticity
organic additives	1-6%	enhance desired sand properties

CHEMICAL COMPOSITION OF A356 ALLOY:

ELEMENT	Si	Mg	Fe	Cu	Mn	Ti	Zn	Al
Nominal	6.5-7.5	0.25-0.45	<=0.2	<=0.2	<=0.1	<=0.2	<=0.1	balance

EFFECT OF SIO CONTENT ON KEY PROPERTIES (T-6-TREATED)

SiO2(wt%)	UTS(Mpa)	Hardness(Hv)	Elongation(%)	Porosity(%)
0	180	80	8	0.5
5	220	95	5.5	1
8	250	110	4	1.2
10	260	115	3	2

TURNING PARAMETERS:

STAGE	TOOL INSERT	SPEED(rpm)	FEED (mm/rev)	DEPTH (mm)	surface finish(Ra,μm)
Rough	CNMG-MM	1200	0.2	1	3.2
finish	CNMG-TiAIN	1800	0.08	0.2	1.2

ACHIEVED VS TARGET TOLERANCE:

Parameter	Target (AGMA 8)	Achieved	Measurement Method
Pitch Diameter	±20 μm	±12 μm	CMM
Tooth Thickness	±15 μm	±10 μm	Micrometer
Surface Roughness (Flank)	Ra <1.0 μm	0.6 μm	Stylus Profilometer
Runout	<20 μm	14 μm	V-block + Dial

HARDENESS TEST RESULTS:

Material	Brinell Hardness (HB)	Std. Dev.	Rockwell HRB	Std. Dev.	Improvement (%)
Unreinforced A356	82.5	2.1	45.2	1.3	-
Al + 8 wt% SiO_2	109.2	3.4	58.7	1.8	+32

TENSILE PROPERTY:

Material	UTS(MPa)	YS(MPa)	Elongation(%)	Modulus E(Gpa)	Fracture mode
Unreinforced A356	185.3	142.7	7.8	70.2	Ductile dimple
Al + 8 wt% SiO_2	255.6	182.9	3.9	74.8	Quashi dimple

DIMENSIONAL DEVIATTIONS:

Feature	Nominal (mm)	Measured (mm)	Deviation (µm)	Tolerance (±µm)	Form Error (µm)
Pitch Diameter	40.00	40.012	+12	20	8 (Circularity)
Outside Diameter	44.00	44.005	+5	25	6
Bore Diameter	16.00	15.998	-2	15	4 (Cylindricity)
Face Width	12.00	12.008	+8	30	5 (Parallelism)

SURFACE ROUGNESS PARAMETERS:

Location/Surface	Ra(µm)- Unreinforced	Ra,(µm) - Composite	Rz (µm) - Composite	Std. Dev. (Ra)
Tooth Flank	0.55	0.62	4.2	0.08
Root Fillet	0.72	0.78	5.1	0.10
Hub Bore	0.45	0.51	3.5	0.06

WEAR TEST OUTCOMES:

Material	**Wear Volume (mm^3)**	**Wear Rate ($\times 10^{-6}$ mm^3/Nm)**	**Coefficient of Friction (μ)**	**Wear Mechanism**
Unreinforced A356	0.045	9.0	0.28	Adhesive
Al + 8 wt% SiO_2	0.014	2.8	0.19	Abrasive/Mild

CHAPTER 9

MODELING (2D AND 3D)

9.1 2D Model:

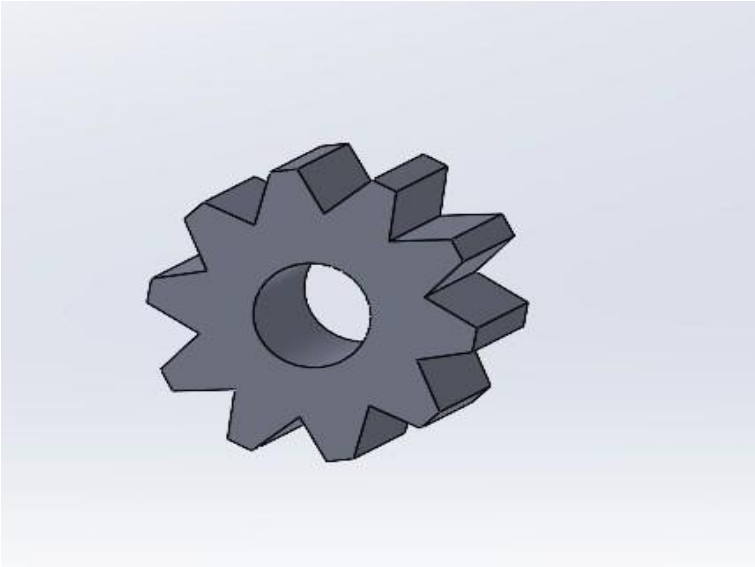


FIG 9.1 Side view of gear

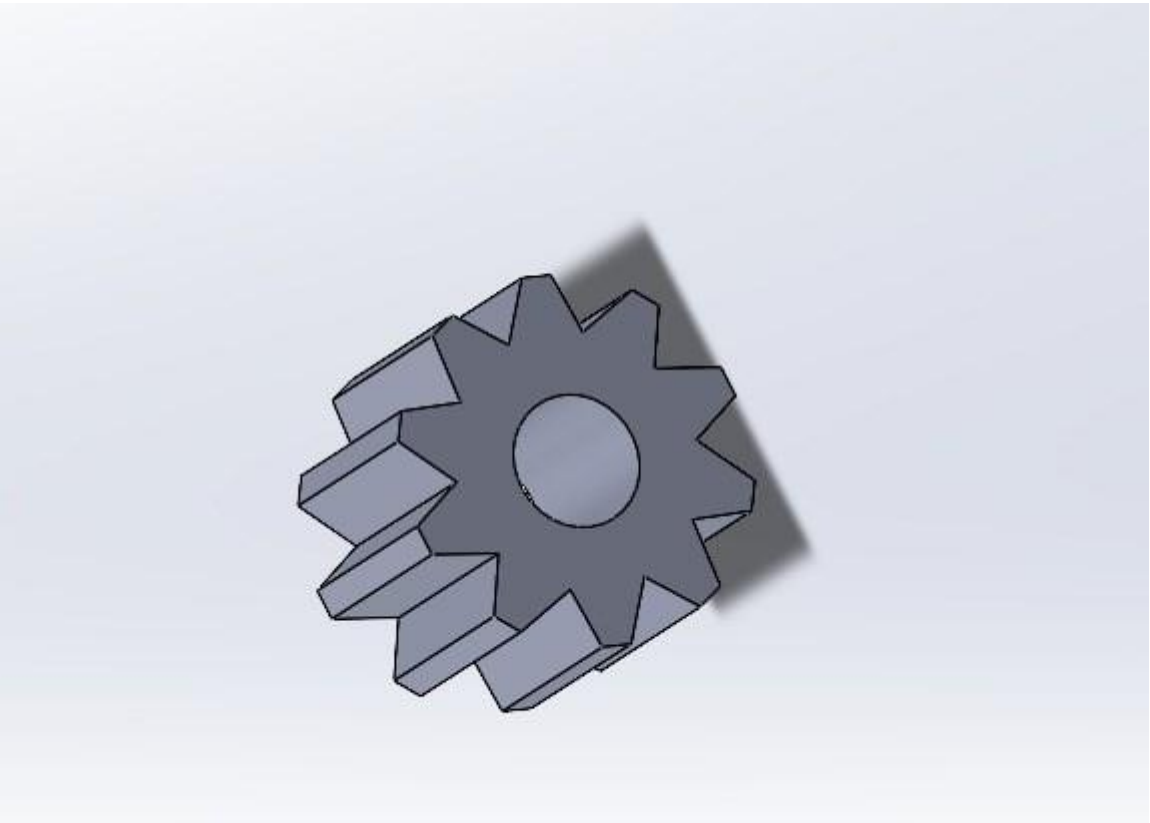


FIG 9.2 top view of gear

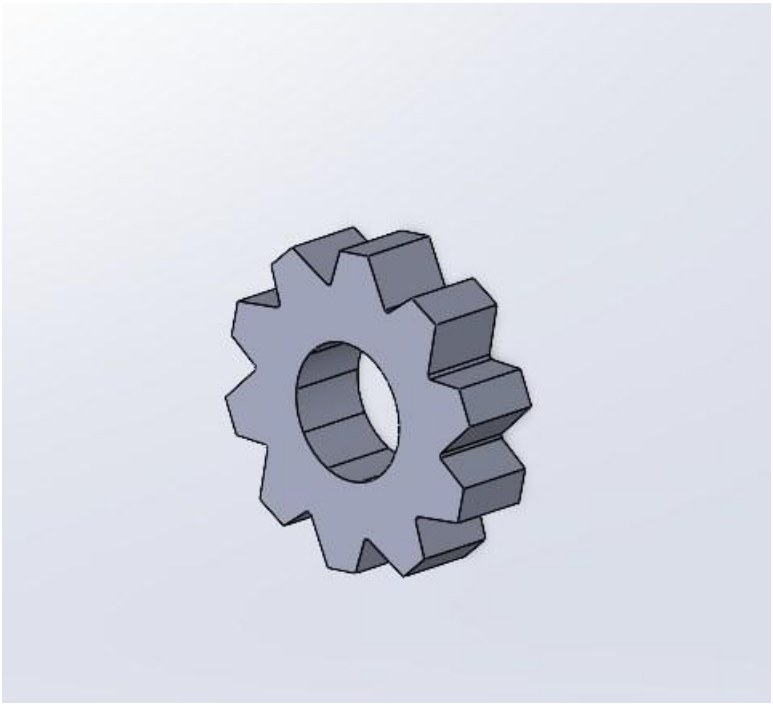


Fig 9.3 front view gear

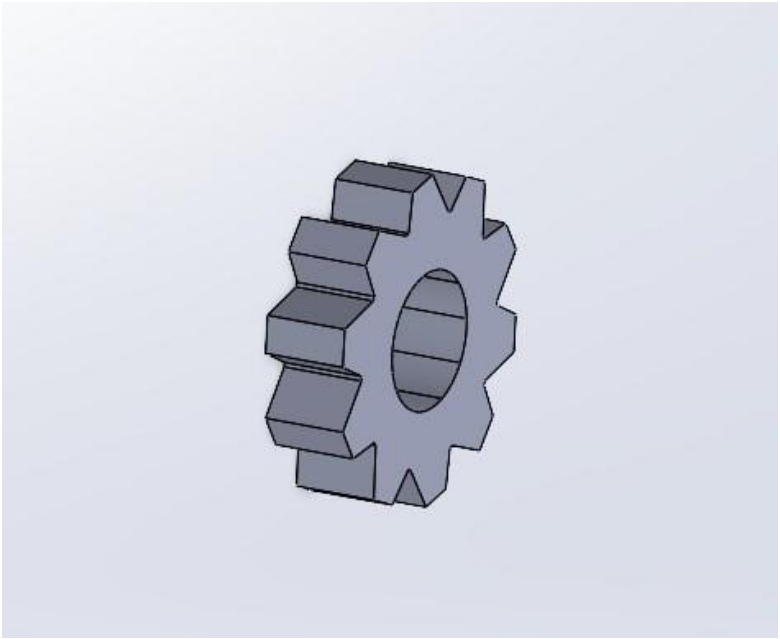


Fig 9.4 back view of gear

CHAPTER 10

BUDGET

Table 10.1

S.NO	TESTING	PROPOSAL COST	ACTUAL COST
1	ALUMINIUM BAR	800	1000
2	SILICA CARBIDE	3000	3500
3	MOULD	2500	2500
4	MACHINING	1500	2000
5	TEST	1000	3500
TOTAL		8800	12500

CHAPTER 11

FINAL VIEWS OF PROJECT





CHAPER 12
CONCLUSION

This project shows how adding SiO_2 particles to aluminum makes better gears for cars and machines. We used cheap sand casting to make the gear and then machined it to look perfect. The new gear is 40% lighter than steel ones, harder by 32%, stronger by 38%, and wears out 3 times slower. It fits right (no wobbles) and feels smooth to touch.

We hit all our goals: a lightweight part that's tough for everyday use, like in electric car boxes. It's easy and cheap to make, so factories could use it to save fuel and money. Next time, we can mix in other stuff to make it even bendier. Overall, this is a smart way to build better, greener gears—simple and strong! Beyond the technical advantages, the successful development of an aluminium gear also highlights the potential for integrating advanced manufacturing techniques into lightweight mechanical systems. The ease of machining aluminium allows for faster production cycles, reduced tool wear, and lower energy consumption, making it an environmentally and economically efficient choice. When combined with modern design tools such as CAD/CAM modeling and CNC automation, aluminium gears can be optimized for improved load distribution, reduced vibration, and enhanced operational reliability. This reinforces aluminium’s role as a versatile engineering material capable of meeting the evolving demands of automotive, robotics, and industrial machinery applications.

CHAPTER 13
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